

# Aditya Dasika

AI/ML Engineer · Agentic & Multi-Agent Systems · RAG · LLM Eval, Observability & Guardrails

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## EXPERIENCE

### AI/LLM Engineer II

*Phenom People*

April 2025 – Present

*Hyderabad, Telangana*

- Architected **chatbotCXA**gent, a production multi-agent system on **LangGraph** (12 orchestrated nodes) spanning intent recognition, planning, dialogue-state management, and tool routing — with **Graph RAG** retrieval over enterprise knowledge and document repositories via FAISS vector search and a Neo4j graph for multi-hop, policy-aware grounding.
- Engineered tool-calling agents that execute governed actions — database queries and structured extraction from unstructured documents — under **XGrammar constrained decoding** and schema-validated JSON guardrails, raising tool-routing accuracy to **91.9%** and eliminating malformed tool calls in live traffic.
- Building the detection and decisioning layer of an enterprise **hiring-integrity / fraud-risk platform** — an **LLM orchestration engine** that investigates flagged candidate activity, correlates evidence across signals, and produces reviewer-facing assessments with supporting rationale — backed by a **policy-driven rules engine** (score / condition → action), per-tenant risk configuration, and audited review workflows across the signal lifecycle.
- Built an **LLM evaluation framework** (pass@k, LLM-as-judge rubric) gating every release across all 12 nodes, instrumented with **LiteLLM + Langfuse** for per-node cost, latency, and full-trace auditability — continuously monitoring agent behaviour and curbing hallucinations.
- Fine-tuned and served Qwen models via **QLoRA** (Unsloth, NF4) on **vLLM** / AWS g5.xlarge (A10G), sustaining 10.98 RPS at 98.6% success under load at roughly half the inference cost of hosted GPT-4.1.
- Fine-tuned a multi-task **ModernBERT** (intent + NER) in PyTorch, raising NLU accuracy from 87% to 97%.

### Full Stack Engineer

*Blueleaves Farms*

May 2022 – February 2025

*Hyderabad, Telangana*

- Built data-driven analytics on large-scale transactional and stock-movement data — demand forecasting, sales-trend, and inventory models that surfaced discrepancies and improved procurement/replenishment decisions across 5+ locations and 10,000+ monthly transactions.
- Owned end-to-end delivery of production systems (POS, CRM, inventory, procurement, e-commerce) on Node.js, Firebase Cloud Functions, and REST APIs — with role-based access control and CI/CD — integrating multiple internal systems and cutting manual operational errors by roughly 90%.

## PROJECTS

### NorthStar — Motorcycle Navigation Companion App | *Android, BLE, Maps SDK*

- Built and shipped a companion app for the Royal Enfield Tripper Dash, delivering turn-by-turn navigation to the bike's connected instrument cluster and growing an organic user base without paid distribution.
- Traction drew **Royal Enfield's** attention directly, leading to an on-site visit at their HQ and a continuing relationship — they have since approached him about a role and about collaborating on other motorcycle-accessory work.

### IndexNotes — AI Knowledge Graph for Scientific Papers | *Neo4j, Ollama, PyMuPDF, Python, Docker*

- Built an automated Graph RAG pipeline ingesting research papers (PDF → local-LLM extraction → structured JSON → Neo4j), creating 40+ concepts and 17+ typed relationships, with quantized local models via Ollama for entity/relationship extraction.
- Designed a Neo4j schema enabling multi-hop graph traversal and cross-document retrieval across scientific domains.

## TECHNICAL SKILLS

**Languages:** Python, JavaScript, TypeScript, SQL, Bash

**GenAI & Agentic:** Agentic / Multi-Agent Systems, LangGraph, LangChain, Tool/Function Calling, Prompt Engineering, Model Routing, Context-Window Management, RAG & Graph RAG, LLM Fine-Tuning (QLoRA, SFT)

**Eval, Observability & Guardrails:** LLM Evaluation (LLM-as-judge, pass@k), OpenTelemetry, Constrained Decoding (XGrammar), Schema Validation, Langfuse, LiteLLM, Model Monitoring, Hallucination Mitigation, Role-Based Access Control

**ML & Serving:** PyTorch, Transformers (BERT/ModernBERT), HuggingFace PEFT, Unsloth, Physics-Informed Neural Networks, vLLM, Embeddings / Vector Search (FAISS), Neo4j, Ollama, Production MLOps

**Backend & Infra:** FastAPI, Node.js, Express.js, REST APIs, Microservices, Docker, Helm, Temporal, Kafka, AWS (EC2/S3/SageMaker), GitHub Actions, CI/CD, Linux

**Frontend & Auth:** React, Vite, Micro-Frontends, Keycloak, OAuth / SSO, Design Systems, Role-Based Access Control

**Databases:** PostgreSQL, Neo4j, MongoDB, ClickHouse, Redis, Firebase

## EDUCATION

**Andhra Loyola Institute of Engineering and Technology**

*Bachelor of Technology in Mechanical Engineering*

Vijayawada, Andhra Pradesh

*August 2018 – May 2022*